

NAME

mbpy_histplot – Create an executable shellscript which will generate a GMT modern-mode frequency histogram plot of a set of data values.

VERSION

Version 5.0

SYNOPSIS

```
mbpy_histplot -Ifile [-Oroot] [--formatfmt] [--titletext] [--xlabeltext] [--ylabeltext] [--dpidpi] [-X]
[-V]
[-C[xmin/xmax/]width] [-Gcolor] [-Wpen]
[-Rxmin/xmax/ymin/ymax] [-Jspec] [-Ppagesize] [--portrait | --landscape] [--tick-infospec]
[--gmt-defaultparam/value]
```

DESCRIPTION

mbpy_histplot is a Python/PyGMT-era replacement for the Perl macro **mbm_histplot**. It reads a file containing one data value per line and generates an executable shellscript of **GMT modern mode** commands, built around **gmt histogram** (the modern-mode name for classic **pshistogram**), which produces a frequency histogram of those values when run: the bars show what percentage of the data falls in each value bin, and mean / ± 1 standard deviation / ± 2 standard deviation normal-distribution curves are always overlaid (in black, red, and green respectively) for reference.

mbpy_histplot shares its page-layout and projection-fitting conventions with **mbpy_grdplot** and **mbpy_grd3dplot** (and, in fact, the underlying **mbpy_layout.py** module), except that its default projection is a plain linear/Cartesian one (matching **mbm_histplot**'s own default of **-Jx**, rather than the Mercator/auto-fit default the grid-plotting macros use) -- appropriate since a histogram's axes are ordinary numeric value/percentage scales, not geographic coordinates.

Unlike **mbm_histplot**'s **-Ltitle:xlabel:ylabel** colon-joined label option, **mbpy_histplot** exposes **--title/--xlabel/--ylabel** as separate options.

mbpy_histplot does not expose a data-rescaling, xy-overlay, or custom CPT-file option: **mbm_histplot** itself refers to variables for all of these (**\$data_scale**, **@xyfiles**, **\$file_cpt**) that are never actually assigned from any command-line option in its own source -- they are leftover, unreachable copy-paste from the **mbm_grdplot** template **mbm_histplot** was originally cloned from, not real functionality.

Note: classic **pshistogram**'s outline-pen option was **-Lpen**; the current **gmt histogram** module (checked directly against its own built-in help) has no **-L** option at all and uses **-Wpen** instead, like nearly every other GMT plotting module's pen option. **mbpy_histplot**'s own **-W/ --pen** option is written out to the generated script using the current **-W** syntax, not the classic **-L** **mbm_histplot**'s own Perl source still emits (which is simply not accepted by a current GMT installation).

mbpy_histplot requires **GMT 6** or later (modern mode) and a Python 3 interpreter. It depends on **mbpy_layout.py** (shared with **mbpy_grdplot/mbpy_grd3dplot**) and its own **mbpy_histplot_script.py**, installed alongside it.

MB-SYSTEM AUTHORSHIP

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OPTIONS

- I, --input *file***
Input data file, one numeric value per line. Required.
- O, --output *root***
Output root name. Default: the input file's own basename.
- format *fmt***
Output format passed to **gmt figure**. Default **pdf**.
- title *text***
Plot title. Default: "Frequency Histogram of *file*".
- xlabel *text***
X-axis (value) label. Default: none.
- ylabel *text***
Y-axis (frequency) label. Default: "Frequency %".
- dpi *dpi***
The final raster's resolution (via **gmt figure**).
- X, --execute**
Run the generated script immediately after writing it.
- V, --verbose**
Print the generated script's path and a one-line summary.
- C, --cell-width [*xmin/xmax/*]*width***
Histogram bin width. If given as three slash-separated values, also overrides the data's own detected x (value) bounds. Default width: (xmax-xmin)/10.
- G, --fill *color***
Bar fill color, or **N** for unfilled (outline-only) bars. Default **gray**.
- W, --pen *pen***
Bar outline pen. Default **1p**.
- R, --region *xmin/xmax/ymin/ymax***
Plot region. Default: the data's own value range for x; a y (frequency-percent) range computed from a pre-scan of the data, with a 10% margin above the tallest bin.
- J, --projection *spec***
GMT projection/scale spec, e.g. **x0.02** (linear, 0.02 inch/unit). Default: an auto-fit linear (Cartesian) projection chosen to fill the page.
- P, --pagesize *size***
Page size code. Default **a**.
- portrait, --landscape**
Force the page orientation instead of choosing automatically.
- tick-info *spec***
Raw **-B** basemap tick-interval spec, overriding the automatically computed one.
- gmt-default *param/value***
Override a GMT default parameter for this plot; repeatable.

EXAMPLES

A histogram of a column of survey depth values, with custom bin width, fill color, and labels, rendered directly to a PNG:

```
mbpy_histplot -I depths.txt -O depth_hist \  
  --cell-width 2.5 --fill orange --pen "1p,black" \  
  --title "Depth Distribution" --xlabel "Depth (m)" \  
  --format png --execute -V
```

An outline-only (unfilled) histogram with an explicit bin range:

```
mbpy_histplot -I values.txt -O values_hist \  
  --cell-width "0/100/5" --fill N --pen "2p,blue" \  
  --format pdf -V
```

SEE ALSO

mbm_histplot(1), **mbpy_grdplot(1)**, **mbpy_grd3dplot(1)**, **mbpy_xyplot(1)**, **mbhistogram(1)**, **mbdefaults(1)**, **mbsystem(1)**

BUGS

Please report any bugs to the MB-System project on GitHub (<https://github.com/dwcaress/MB-System>).